

1 Questions

- Given the curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$, determine:
 - The area under the curve on the given interval bounded by the x -axis.
 - If the area in (a) is revolved about the x -axis, determine the resulting volume.
 - Determine the centroid of the solid in (b) without using Pappus' theorem.
 - If the interval $-1 \leq x \leq 1$ is removed, determine the arc length of the curve in first quadrant, without using integration.
 - Based on the arc obtained in (d), if it is revolved about the y -axis, determine the centroid of the resulting surface area using Pappus' theorem.
- Using integration, determine the arc length of the curve $y = 5x$ on the interval $0 \leq x \leq 1$.

2 Solutions

- (a) The area under the curve can be calculated using the formula

$$A = \int_{-1}^1 \sqrt{4 - x^2} dx$$

Let $x = 2 \sin \theta$, then $dx = 2 \cos \theta d\theta$ and the limits of integration change to $\theta = -\pi/6$ and $\theta = \pi/6$.
Thus,

$$\begin{aligned}
 A &= \int_{-\pi/6}^{\pi/6} \sqrt{4 - 4 \sin^2 \theta} \cdot 2 \cos \theta d\theta \\
 &= \int_{-\pi/6}^{\pi/6} 4 \cos^2 \theta d\theta \\
 &= 4 \int_{-\pi/6}^{\pi/6} \cos^2 \theta d\theta \\
 &= 4 \int_{-\pi/6}^{\pi/6} \frac{1 + \cos(2\theta)}{2} d\theta \\
 &= 2 \int_{-\pi/6}^{\pi/6} (1 + \cos(2\theta)) d\theta \\
 &= 2 \left[\theta + \frac{\sin(2\theta)}{2} \right]_{-\pi/6}^{\pi/6} \\
 &= 2 \left[\frac{\pi}{6} + \frac{\sin(\pi/3)}{2} - \left(-\frac{\pi}{6} + \frac{\sin(-\pi/3)}{2} \right) \right] \\
 &= 2 \left[\frac{\pi}{6} + \frac{\sqrt{3}}{4} + \frac{\pi}{6} + \frac{\sqrt{3}}{4} \right] \\
 &= 2 \left[\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right] \\
 &= \frac{2\pi}{3} + \sqrt{3}
 \end{aligned}$$

- The volume of the solid obtained by revolving the area under the curve about the x -axis can be

calculated using the formula

$$\begin{aligned}
 V &= \pi \int_{-1}^1 (\sqrt{4-x^2})^2 dx \\
 &= \pi \int_{-1}^1 (4-x^2) dx \\
 &= \pi \left[4x - \frac{x^3}{3} \right]_{-1}^1 \\
 &= \pi \left[\left(4 - \frac{1}{3} \right) - \left(-4 + \frac{1}{3} \right) \right] \\
 &= \pi \left[\frac{11}{3} + \frac{11}{3} \right] \\
 &= \frac{22\pi}{3}
 \end{aligned}$$

(c) The centroid of the solid can be calculated using the formula

$$\begin{aligned}
 \bar{y} &= \frac{\frac{1}{2} \int_a^b [f(x)]^2 dx}{\int_a^b f(x) dx} \\
 &= \frac{\frac{1}{2} \int_{-1}^1 (\sqrt{4-x^2})^2 dx}{\int_{-1}^1 \sqrt{4-x^2} dx} \\
 &= \frac{\frac{1}{2} \int_{-1}^1 (4-x^2) dx}{\frac{2\pi}{3} + \sqrt{3}} \\
 &= \frac{\frac{1}{2} \cdot \frac{22\pi}{3}}{\frac{2\pi}{3} + \sqrt{3}} \\
 &= \frac{11\pi}{3} \cdot \frac{1}{\frac{2\pi}{3} + \sqrt{3}} \\
 &= \frac{11\pi}{3} \cdot \frac{3}{2\pi + 3\sqrt{3}} \\
 &= \frac{11\pi}{2\pi + 3\sqrt{3}}
 \end{aligned}$$

Since the solid is symmetric about the y -axis, then $\bar{x} = 0$. Thus, the centroid of the solid is at $\left(0, \frac{11\pi}{2\pi+3\sqrt{3}} \right)$.

(d) The arc length of the curve in the first quadrant is one quarter of the circumference of a circle with radius 2, which can be calculated using the formula

$$K = \frac{1}{4} \cdot 2\pi r = \frac{1}{4} \cdot 2\pi \cdot 2 = \pi$$

(e) The centroid of the surface area obtained by revolving the arc about the y -axis can be calculated using Pappus' theorem, which states that the centroid of a surface of revolution is located at a distance from the axis of revolution equal to the distance traveled by the centroid of the generating curve. Since the centroid of the arc is located at a distance of $\frac{4}{\pi}$ from the y -axis, then the centroid of the surface area is located at a distance of $\pi \cdot \frac{4}{\pi} = 4$ from the y -axis. Thus, the centroid of the surface area is at $(4, 0)$.

2. The arc length of the curve can be calculated using the formula

$$\begin{aligned} K &= \int_0^1 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_0^1 \sqrt{1 + 25} dx \\ &= \int_0^1 \sqrt{26} dx \\ &= \sqrt{26} \end{aligned}$$